

Working Principle & System Flow

How the Automatic Exhaust Fan detects smoke and responds

Overview

The Automatic Exhaust Fan system continuously monitors the air quality using the MQ-2 gas sensor. When smoke or harmful gas is detected above a set threshold, the system immediately activates the exhaust fan, sounds a buzzer alarm, and turns on the red hazard LED — all without any manual intervention. When the air is clean, the green LED glows and the fan stays off, conserving energy.

Step-by-Step Working

Step 1 — Power ON	The ESP8266 NodeMCU powers up and initializes all pins. The MQ-2 sensor begins warming up (~20 seconds).
Step 2 — Continuous Monitoring	The loop() function runs continuously. analogRead(A0) reads the MQ-2 sensor output every 1 second.
Step 3a — Smoke Detected	Red LED turns ON. Buzzer sounds the alarm. Relay activates → Exhaust Fan turns ON. Serial monitor shows "Smoke Detected".
Step 3b — Clean Air (level < 500)	Green LED turns ON. Buzzer is deactivated. Relay deactivates → Exhaust Fan turns OFF. Serial monitor shows "Air Clean".
Step 4 — Repeat	The system loops back to Step 2 and repeats every 1 second.

LED Indicator Summary

LED Color	Pin	Condition	Meaning
■ Green	D6	smokeLevel < 500	Air is clean — Fan OFF
■ Red	D3	smokeLevel > 500	Smoke detected — Fan ON, Buzzer ON
■ Orange	D7	smokeLevel = 500	At threshold — Fan ON (warning)

Energy Efficiency Benefit

Traditional exhaust fans run continuously regardless of air quality. This system runs the fan ONLY when smoke is detected, saving electricity and reducing motor wear. This also significantly reduces noise in the environment since the fan is silent when not needed.